3.4.2 Mechanical properties of matter

	Learning outcomes	Additional guidance
	Learners should be able to demonstrate and apply their knowledge and understanding of:	
(a)	force–extension (or compression) graph; work done is area under graph	*M3.1*
(b)	elastic potential energy; $E = \frac{1}{2}Fx$; $E = \frac{1}{2}kx^2$	*M0.5, M3.12*
(c)	stress, strain and ultimate tensile strength	
(d)	**(i)** Young modulus = $\frac{\text{tensile stress}}{\text{tensile strain}}$, $E = \frac{\sigma}{\varepsilon}$	*M3.1*
	(ii) techniques and procedures used to determine the Young modulus for a metal	**PAG2**
(e)	stress–strain graphs for typical ductile, brittle and polymeric materials	*M3.2* HSW8
(f)	elastic and plastic deformations of materials.	HSW4, 5, 9, 12 Investigating the properties of materials **PAG2**